

KRISHNA VAMSI PRATHIPATI

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Profile

Engineer focused on information retrieval, ranking systems, and scalable ML pipelines, with experience building multi-stage retrieval and reranking workflows grounded in structured data and citations. Designed systems with metadata-aware ranking, adversarial validation, and dual-pass retrieval strategies to improve search relevance and reliability, supported by a strong foundation in deep learning, linear algebra, and optimization. Experienced in developing low-latency, end-to-end pipelines for real-time inference and deployment using Python and APIs, with a growing interest in solving large-scale search, retrieval evaluation, and efficiency challenges in production systems.

Education

Velagapudi Ramakrishna Siddhartha Engineering College Vijayawada, India
Bachelor of Technology in Artificial Intelligence and Data Science; CGPA: 8.85 Nov 2022 – May 2026

RESEARCH & PUBLICATIONS

- Detection of Aircraft from SAR Images Using Deep Neural Networks** [GitHub](#) 
- Authored and published paper using enhanced YOLOv8 + GLCM/CLAHE on TerraSAR-X dataset (844 images); achieved 95.3% accuracy, 93% precision, 91% F1-score in noisy environments.
 - Replaced color features with texture analysis for robustness against weather/urban noise.
 - DOI: [10.1109/INSPECT63485.2024.10896234](https://doi.org/10.1109/INSPECT63485.2024.10896234) 

Projects

ScholarGraph Elite — Adversarial Agentic Research System

- Built a multi-stage retrieval pipeline using LangGraph with 4 agents: planning, retrieval, validation, and reranking.
- Implemented dual-pass search strategy: Relevance-ranked retrieval, Citation-graph-informed retrieval
- Designed custom ranking function (Authority Weightage) combining: Abstract depth (40%), Recency (35%), Author signals (15%), Title specificity (10%)
- Developed an adversarial evaluation loop where outputs are recursively rejected and re-ranked if below quality threshold
- Effectively simulates: Search comprehensiveness, Result reliability, Citation-grounded generatio
- Closely mirrors IR benchmarking + reranking problems in production systems

Face recognition using Vision Transformers (ViT)

[GitHub](#) 

- Designed low-latency pipeline combining Haar cascades + ViT embeddings.
- Achieved 98.24% accuracy on 2,100-image dataset.
- Optimized ROI extraction + inference efficiency, reducing compute overhead.

Train Wagon Anomaly Detection

[GitHub](#) 

- Built low-latency detection system using YOLOv8 + OpenCV.
- Achieved 2s inference latency with real-time alerts (Telegram integration).
- Focused on pipeline optimization, streaming input, and deployment reliability.

Behavioral change monitoring software

- Built Flask dashboard with GPT-4/NLTK for real-time multi-class sentiment analysis; optimized preprocessing to cut API costs, visualized trends in Matplotlib/Seaborn.
- Optimized preprocessing to reduce API cost + latency.

- Designed structured outputs + visualization pipelines.

Technical Skills

Information Retrieval & Search Systems:

- Multi-stage retrieval pipelines (relevance + metadata-aware reranking).
- Ranking signal design using structured features (recency, authority, semantic depth).
- Query decomposition and dual-pass retrieval strategies.
- Citation-aware retrieval and grounded generation (project-based).

Mathematics & Optimization:

- Linear Algebra (vectors, similarity measures, embeddings).
- Optimization techniques for model performance and pipeline efficiency.
- Understanding of distance metrics and ranking relevance (cosine similarity, feature weighting).

Machine Learning & Deep Learning:

- PyTorch, TensorFlow, Transformer architectures (Vision Transformers).
- Feature engineering for noisy datasets (GLCM, CLAHE).
- Model optimization for accuracy-latency trade-offs.
- Experience with real-time inference pipelines.

Programming & systems:

- Python (NumPy, Pandas, scikit-learn), SQL.
- Strong problem-solving and algorithmic thinking.
- API design using Flask and REST.
- Git for version control.

Generative AI & NLP:

- LLM API integration (GPT-based systems).
- Prompt engineering and structured output design.
- Retrieval-augmented workflows.

Certifications & Accomplishments

- Achieved Finalist position in the Smart India Hackathon 2023, a national hackathon conducted by the Indian government.
- **Certifications:** Infosys AI/Deep Learning, Oracle Database.
- Volunteered with Make A Difference to teach Mathematics to students in Grades 5 and 7, contributing to their academic growth through structured lessons and mentorship.